Lynq NTN-Pulse

Every signal, the right path.

CHALLENGE	OQ Technology — AnyPoint-NTN: Global gNB-Onboard LEO Constellation for Ubiquitous 5G NR-NTN
CATEGORY	Next-Gen Downstream
TEAM	Lynq — Davoud Jokar, Ines Bouissou, Gambhir Sharma, Maryam Amini

01 Executive Summary

Lynq NTN-Pulse is a software-defined digital twin and policy engine for OQ Technology's regenerative gNB-onboard LEO constellation. It governs industrial traffic across onboard gNB, feeder/ISL paths, gateways, 5G Core and cloud applications. A 3GPP-aligned model applies policies for QoS, latency, availability, data residency, spectrum and cost.

DESIGN POINT

~12	10°	100%	RF+optical
satellites in orbit today (OQ's current LEO fleet)	minimum elevation	full regenerative gNB on every satellite	RF intra-plane ISL, optical cross-plane relay subset

—— Continuous	●—● Mesh-relayed	··· Store-and-forward
<150 ms	<250 ms	≤90 min

The first beachhead is offshore energy and maritime operations. Lynq adds the software layer that makes regenerative NTN capacity measurable and deployable.

02 Problem & Opportunity

Status quo: the bent-pipe limit. Traditional transparent payloads relay radio signals but require a visible ground gateway and feeder link for network processing, routing and terrestrial-core access. If the gateway is outside the footprint, geographically blocked, nationally restricted or overloaded, service degrades through delay, inefficient routing or loss of connectivity.

This affects offshore wind farms, energy platforms, vessels and pipelines. Their traffic needs different treatment: alarms and control commands require priority, while telemetry, video and bulk files can tolerate delay.

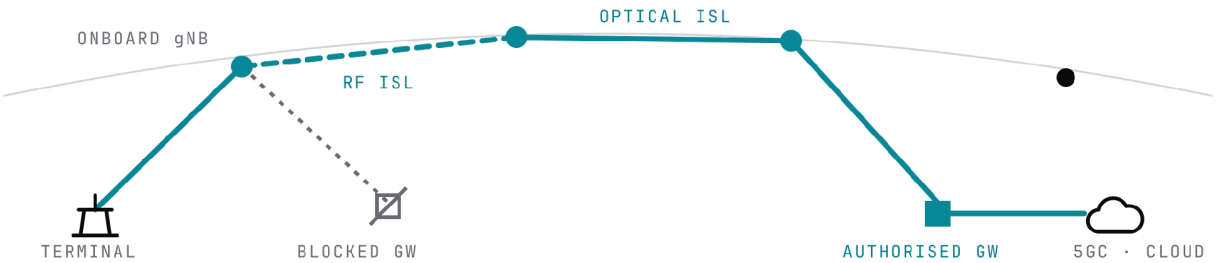
A regenerative gNB moves selected RAN and Layer 2/3 functions into orbit, but introduces Doppler and delay, handovers, intermittent feeder access, gateway selection, ISL routing and onboard compute limits. Lynq provides the policy layer that decides whether traffic is delivered immediately, routed, edge-processed or stored for later downlink.

03 Proposed Direction

Technical angle

Lynq combines:

- **Digital twin:** models satellite position, coverage, gateway visibility, feeder/ISL state, terminal location, traffic demand and onboard compute.
- **Policy engine:** assigns traffic classes to available paths using QoS, latency, reliability, residency and cost.
- **NTN model and dashboard:** covers timing, beam management, handover, link budgets, scheduling, queues, predicted outages and route decisions.



Data path: industrial terminal → regenerative onboard gNB → policy engine → ISL/feeder → authorised gateway and 5G Core → enterprise application. Teal = policy-compliant path; grey = blocked gateway.

Constellation (sized with the twin). We start with OQ's 12 LEO satellites and focus them on one region: the North Sea offshore energy basin (UK, Norwegian, Danish, Dutch and German sectors, ~51–61°N). At a typical ~500–550 km altitude and a 10° minimum elevation, one footprint is ~3,000 km across, so a single satellite in view covers the whole basin and every land gateway around it. Twelve satellites give frequent passes rather than continuous coverage: alarms go out on the next pass and store-and-forward carries

bulk data between passes. At a 5-year life, that is ~2–3 replacements per year. The design can later grow with more satellites to close the gaps between passes and extend to the Norwegian Sea and Atlantic approaches.

Payload trade. Full regenerative gNB: bent-pipe cannot serve users with no gateway in view, and split gNB (options 2/7.2x) keeps most RF/PHY power onboard while RRC still round-trips over the feeder. Power, thermal and radiation costs are mitigated by a narrowband gNB, duty-cycling over empty ocean and COTS processing under a rad-hard supervisor. The onboard gNB also sorts every message into one of three classes, which is why the scheduler must sit in orbit: **critica**l (alarms, safety and shutdown commands, a few hundred bytes) takes the first uplink slot and the first authorised path on the current pass; **regular** (sensor telemetry, logs, files) is queued onboard and delivered by store-and-forward on a later pass (≤90 min); **metadata** (device status, position, delivery receipts and policy tags) is tiny but tells the twin and the policy engine where each asset is and which gateway it may use. A bent-pipe payload cannot make this call without a gateway in view.

Example: mid-Atlantic platform → Frankfurt cloud in `~75–110 ms` one-way over ISL, versus no service on bent-pipe. The design uses 3GPP Rel-17 as the NTN baseline, incorporates Rel-18 enhancements and keeps a Rel-19-ready roadmap.

It evaluates ISL topology (optical/laser vs. RF Ka/V/W-band): optical links offer high capacity and low RF interference but require precise acquisition, pointing and tracking; RF links provide different acquisition and availability characteristics but add spectrum, antenna and interference constraints. Narrowband traffic needs only single-digit Mbps per satellite, so we choose RF intra-plane ISL on all satellites (near-static geometry, proven by Iridium) and optical cross-plane on a relay subset (no spectrum licence, hard to intercept), from European suppliers to avoid ITAR/EAR.

The system implements advanced Doppler/delay pre-compensation from moving platforms using applicable 3GPP Rel-17/18/19 procedures. AI supports traffic forecasting, anomaly detection and resource prioritisation; deterministic timing, link-budget and safety controls remain authoritative.

Regulatory angle

The Consultant's Dilemma is that a local ground gateway may be technically attractive but unavailable because of national restrictions, data-residency rules or licensing. A regenerative gNB knows the user's identity, PLMN and verified location, so Lynq applies routing policy onboard and sends each flow through an authorised ISL/feeder path only to a policy-compliant gateway with local user-plane breakout. Under MNO partnerships, the MNO keeps lawful intercept on its own core.

The compliance layer treats ITU Radio Regulations Article 9 coordination and Article 22 EPFD limits as hard design constraints (feeder-link masks, gateway exclusion zones), together with applicable PFD/EIRP conditions and national authorisations. It supports EU-hosted processing, encryption, key ownership, data minimisation, pseudonymisation and controlled GDPR retention. **Sovereignty is built into routing.**

Engagement sequence

01	02	03	04	05
Luxembourg ILR/LSA Extend OQ's Nov-2025 concession to ISL and regenerative operation	**ITU BR via Luxembourg** Priority filings	**International waters** No landing rights needed	**North Sea regulators & MNOs** Coastal coverage	**EC/CEPT** EU 2 GHz MSS, off the critical path

Business angle

The first buyer is a European offshore energy operator or offshore wind O&M provider (e.g., Equinor, Ørsted, or Eni), explicitly aligning with OQ's European funding focus. Lynq reduces blind spots, improves high-priority alarm delivery and lowers dependence on dedicated VSAT and fixed gateways.

Revenue combines integration fees, monthly subscriptions by assets and traffic class, usage charges for data/priority QoS, and MNO wholesale or roaming. For the 12-satellite North Sea network, opex including replenishment is ~€9M/yr: break-even at ~75k offshore assets at €120/yr, or ~9k sovereign/defence endpoints at €1,000+/yr. At ~€2–3/MB we price per asset, never per byte. OQ's EIC Accelerator award (€2.5M grant plus up to €15M equity) for direct-to-mobile connectivity gives a funded route to a second service on the same North Sea network: SOS and safety messaging to unmodified phones for offshore crews and vessels, alongside asset IoT. The business case compares TCO with VSAT, private LTE and manual inspection baselines, benchmarked against Sateliot, Skylo, Starlink D2C/Lynk and Iridium.

04 Key Assumptions & Risks

RISK	ASSUMPTION	MITIGATION
Payload	Final power, compute, radiation and RAN-interface limits may differ.	Use modular, payload-independent policies with cloud/gateway fallback; validate an OQ interface checklist.
ISL / regulatory	Final ISL, gateway and spectrum choices may change.	Simulate optical/RF ISLs, limited gateway visibility and store-and-forward; apply ITU and national rules as hard constraints.
Adoption	Customers may resist terminals, MNO integration or uncertain SLAs.	Start with one wind-turbine alarm use case, measure customer TCO and define a paid pilot with latency, availability and delivery targets.

05 Next Steps

For Phase 2 in Luxembourg, Lynq will simulate one offshore or maritime corridor and compare bent-pipe and regenerative architectures under changing visibility, gateway access, traffic load and ISL conditions. The demonstration will show priority handling,

Doppler/delay compensation, dynamic ISL/feeder/store-and-forward selection, QoS, latency, delivery probability, queue depth, TCO and routing compliance.

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